Lecture 15: Potential Outcomes and RCTs

In this lecture, we will introduce the potential outcomes framework. This is a rigorous framework for thinking about causality, and it will allow us to think about environments in which treatment effects may be different for different individuals. In this lecture, we will use this framework to think rigorously about what we are estimating when we estimate treatment effects with OLS. This will also prepare us for next lecture, where we will think about whose treatment effect is estimated by IV.

Motivation

Suppose that we were estimating the effect of some treatment, Treated_i on an outcome, Y_i. Using the methods that we have discussed in this class so far, we would estimate this with the following regression:

$$Y_i = \alpha + \beta \cdot \text{Treated}_i + \varepsilon_i$$

which we would estimate with OLS.

Note that the way that this regression is written treats β as if it is the same for everyone. In reality, treatment effects vary across individuals. For example:

- Chemotherapy is bad for most people, but is beneficial for those who have cancer.
- College raises your income, but the effect probably varies by major, and perhaps by other attributes.
- An electric vehicle subsidy will cause some marginal people to buy an EV, but has no effect on those who would have bought anyway, or on those who will not buy even with the subsidy.

We next introduce the potential outcomes framework, which allows us to think about these issues formally.

The Potential Outcomes Framework

We will consider a setting, once again, where each individual i can either receive the treatment or not. We denote individual i's treatment status with the indicator, D_i. We have

$$\text{Treatment Status: } D_i = \begin{cases} 1 & \text{if Treated} \\ 0 & \text{if Not Treated} \end{cases}$$

In the potential outcomes framework, each individual i has two **potential outcomes**: an outcome if they receive the treatment, and an outcome if they do not receive the treatment. We denote this as

$$\text{Potential Outcomes: } \begin{cases} Y_i(1) & = \text{Potential Outcome Under Treatment} \\ Y_i(0) & = \text{Potential Outcome Under No Treatment} \end{cases}$$

However, we only actually observe one of the potential outcomes. For treated individuals ($D_i = 1$), we observe $Y_i(1)$, and for untreated individuals ($D_i = 0$), we observe $Y_i(0)$. Equivalently, we can write:

$$\text{Observed Outcome: } Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

We refer to the unobserved potential outcome as the **counterfactual**. The fact that we cannot observe the counterfactual is what makes causal inference challenging, and is sometimes called "the fundamental problem of causal inference."

Notice that in the potential outcomes framework, we wrote each potential outcome as a function of the individual's treatment status, but not of the other individuals' treatment status. This was an implicit assumption: writing things this way rules out spillover effects across units. This assumption is sometimes called either the "Stable Unit Treatment Value Assumption (SUTVA)," or "No Spillovers."

Treatment Effects, Selection Bias, and Randomized Controlled Trials

As economists, we are most often interested in estimating treatment effects: the causal effect of the treatment on the outcome. In the potential outcomes framework, treatment effects can vary across individuals. For each individual, the **treatment effect** is defined as

$$\text{Treatment Effect: } TE_i = Y_i(1) - Y_i(0)$$

Although treatment effects vary across individuals, we will be focused on estimating different averages of treatment effects. The first is the **Average Treatment Effect (ATE)**. This is the average treatment effect across all individuals:

$$\text{Average Treatment Effect: } ATE = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Another interesting average is the **Average Treatment Effect on the Treated (abbreviated ATT or ToT)**. This is the average treatment effect for all individuals that receive the treatment:

$$\text{Average Treatment Effect on the Treated: } ATT = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

What can we estimate in the data? When we estimate treatment effects with OLS, we are simply comparing the mean treated outcome to the mean control outcome. We have:

$$\begin{aligned}
\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] &= \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \\
&= \mathbb{E}[TE_i + Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \\
&= \underbrace{\mathbb{E}[TE_i \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{Selection Bias}}
\end{aligned}$$

When we estimate treatment effects with OLS, we are thus estimating the ATT, plus a selection bias term that reflects the difference in the untreated potential outcome between the treated and untreated groups.[1] Ordinary least squares will give us an unbiased and consistent estimate of the ATT if the selection bias term is zero.

Connection to OLS Orthogonality Condition In the framework above, a comparison of means identifies the ATT if the selection bias term is zero. This is similar to the OLS orthogonality condition: OLS will identify the causal effect, β, if the error term is uncorrelated with the regressor (here, the regressor is the treatment D_i). We can show that these two conditions are not just similar, they are two ways of writing the same thing. If we define β as the ATT, then the orthogonality condition and the zero-selection-bias condition are the same. We will skip the derivation for now, but I have included it at the end of these notes.

Randomized Controlled Trials

Selection bias is a major challenge for the estimation of causal effects. One way to get around selection bias is to run a **randomized controlled trial (RCT)**. In this setting, the researcher controls which individuals are assigned to the treatment and control groups, and assigns individuals to each group randomly. For today, we will assume perfect compliance: the individuals assigned to the treatment actually receive the treatment, and the individuals assigned to the control group do not receive the treatment.

With random assignment, we know that treatment status, D_i, is independent of potential outcomes, $Y_i(1)$ and $Y_i(0)$. With independence, the expectation of potential outcomes will not depend on treatment assignment. The selection bias term is thus zero:

$$\text{With Randomization: } \mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] = \mathbb{E}[Y_i(0)] - \mathbb{E}[Y_i(0)] = 0$$

[1] Traditionally, we write the treatment effect as the ATT, and the selection bias in terms of $Y_i(0)$. However, we could flip things around and write the treatment effect as the average treatment effect on the untreated, in which case the selection bias would be in terms of $Y_i(1)$.

And moreover, the ATT is the same as the ATE:

$$\text{With Randomization: } \underbrace{\mathbb{E}\left[TE_i \mid D_i = 1\right]}_{\text{ATT}} = \underbrace{\mathbb{E}\left[TE_i\right]}_{\text{ATE}}$$

It thus follows that the difference in means identifies the ATT, which is also the ATE.

$$\text{With Randomization: } \mathbb{E}\left[Y_i \mid D_i = 1\right] - \mathbb{E}\left[Y_i \mid D_i = 0\right] = \mathbb{E}\left[TE_i\right]$$

Thus, if we use OLS to estimate treatment effects from an RCT, then our estimator will be unbiased and consistent.

Exercise 1. Consider the following examples of treatment effects. In each case, explain what sort of selection bias might arise if we tried to estimate the treatment effect using observational (not randomized) data:

- The effect of chemotherapy on survival rates
- The effect of a college degree on income
- The effect of military service on life expectancy (*Hint: Note that there is a fitness test to serve in the military!*)

Exercise 2. Suppose that you are studying the effect of union membership on wages. For union members, the average annual salary is \$70,000, and their average counterfactual salary (what they would earn if they were not in a union) is \$60,000. For non-members, the average salary is \$40,000, and their average counterfactual salary (what they would earn if they were in a union) is \$45,000. Answer the following questions:

- Compute the ATT (here, the treatment is union membership).
- Suppose that union members and non-members each make up 50% of the population. Compute the ATE. How do the ATT and ATE compare?
- Compute the difference in observed average salaries between union members and non-members. How does it compare to the ATT? What is the selection bias?

Bonus Material: Estimating Heterogeneous Treatment Effects

We have just shown that if the treatment is randomly assigned, then we can estimate the average treatment effect by comparing the means of the treated group and the control group. Yet treatment effects are potentially heterogeneous across individuals. What can we identify when it comes to heterogeneous treatment effects?

Estimating the Effect of the Treatment on the Distribution

On the one hand, it is straightforward to show how we can estimate the effect of the treatment on the distribution of the outcome. For any value of the outcome, y, we can estimate the effect of the treatment on the cumulative distribution function: the probability that $Y_i \leq y$. To do this, we construct an indicator variable, $1\left(Y_i \leq y\right)$, which is equal to one if Y_i is less than or equal to y, and zero otherwise. We then estimate the regression:

$$1\left(Y_i \leq y\right) = \alpha_y + \beta_y D_i + \varepsilon_{y,i}$$

We can do this for every y that we are interested in, and recover the distributional treatment effects.

Another common approach is to express effects in terms of how the treatment affects the quantiles of the distribution of the outcome: although the estimation procedure for quantile treatment effects is slightly different, it is possible to estimate these as well.

Treatment Effect on the Distribution vs. Distribution of Treatment Effects

However, although we can estimate the effect of the treatment on the distribution, in general **we cannot estimate the distribution of treatment effects**. To see this, imagine that we were interested in the variance of treatment effects. This variance is:

$$\begin{aligned}\text{Var}\left(TE_i\right) &= \text{Var}\left(Y_i\left(1\right) - Y_i\left(0\right)\right) \\ &= \text{Var}\left(Y_i\left(1\right)\right) + \text{Var}\left(Y_i\left(0\right)\right) - 2 \cdot \text{Cov}\left(Y_i\left(1\right), Y_i\left(0\right)\right)\end{aligned}$$

If the treatment is randomized, then we can identify the variances of the potential outcomes:

$$\begin{aligned}\text{With Randomization: Var}\left(Y_i\left(1\right)\right) &= \text{Var}\left(Y_i \mid D_i = 1\right) \\ \text{With Randomization: Var}\left(Y_i\left(0\right)\right) &= \text{Var}\left(Y_i \mid D_i = 0\right)\end{aligned}$$

However, even with random assignment, it will not be possible to identify the covariance of potential outcomes, $\text{Cov}\left(Y_i\left(1\right), Y_i\left(0\right)\right)$. This is because we never observe both potential outcomes: Since we never observe both $Y_i\left(1\right)$ and $Y_i\left(0\right)$, we cannot compute the covariance.

For a specific example, suppose that we have randomized assignment, and we observe $Y_i \sim N\left(0,1\right)$ for the treated group, and also $Y_i \sim N\left(0,1\right)$ for the control group. It is immediate to see that the average treatment effect is zero, since the mean does not change. But there are many possible distributions of the treatment effect. One possibility is that $Y_i\left(1\right) = Y_i\left(0\right)$, and thus $TE_i = 0$, for all i. Another possibility is that $Y_i\left(1\right) = -Y_i\left(0\right)$, and thus $TE_i = -2 \cdot Y_i\left(0\right)$, for all i. Both of these, and many other possibilities, are fully consistent with all of the observable information.

Conditional Average Treatment Effects (CATEs)

Although we cannot estimate the full distribution of treatment effects, it is still possible to estimate heterogeneity in treatment effects across individuals with observably different characteristics. Suppose that for each individual, we observe one or more pre-treatment characteristics, X_i.[2] For example, if we are estimating the effects of a job training program, then X_i might include the individual's age, gender, race, industry, education, and work experience. The **conditional average treatment effect (CATE)** is the average treatment effect for the subset of individuals with particular values of X_i:

$$\text{Conditional Average Treatment Effect: } CATE = \mathbb{E}\left[TE_i \mid X_i = x\right]$$

If we have random assignment of the treatment, then we can estimate the CATE by comparing treated and control observations, subsetting to those individuals with $X_i = x$:

$$\text{With Randomization: } \mathbb{E}\left[Y_i \mid D_i = 1, X_i = x\right] - \mathbb{E}\left[Y_i \mid D_i = 0, X_i = x\right] = \mathbb{E}\left[TE_i \mid X_i = x\right]$$

In practice, for continuous variables it may be difficult to find other observations with the exact same value of X_i. We can then implement this as a regression with interaction effects, although this requires imposing a linear model:

$$Y_i = \alpha + \beta_1 \cdot D_i + \beta_2 \cdot X_i + \gamma \cdot D_i \cdot X_i + \varepsilon_i$$

And then the CATE is $\beta_1 + \gamma \cdot X_i$.

Estimating the conditional average treatment effects will tell us about the component of treatment effect heterogeneity that can be predicted by the observable covariates, X_i. In general, this will give us a lower bound on the total variance of treatment effects: there can also be heterogeneity in treatment effects across individuals who have the same X_i. By the law of total variance:

$$\underbrace{\text{Var}\left(TE_i\right)}_{\text{Total Variance of Treatment Effects}} = \underbrace{\text{Var}\left(\mathbb{E}\left[TE_i \mid X_i\right]\right)}_{\text{Variance of CATEs}} + \underbrace{\mathbb{E}\left[\text{Var}\left(TE_i \mid X_i\right)\right]}_{\text{Average Conditional Variance of Treatment Effects}}$$

[2] The covariates, X_i, need not literally be measured before the treatment occurs. However, we need the covariates to be unaffected by the treatment.

Appendix: Equivalence of Orthogonality and Zero Selection Bias

Here, we will show that if we define β as the ATT, then the OLS orthogonality condition is equivalent to stating that $\mathbb{E}\left[Y_i\left(0\right) \mid D_i = 1\right] - \mathbb{E}\left[Y_i\left(0\right) \mid D_i = 0\right] = 0$.

Define $\beta = \mathbb{E}\left[TE_i \mid D_i = 1\right] = \text{ATT}$. We then have:

$$
\begin{aligned}
Y_i &= \alpha + \beta \cdot D_i + \varepsilon_i \\
\mathbb{E}\left[Y_i\left(0\right) \mid D_i = 0\right] &= \mathbb{E}\left[Y_i \mid D_i = 0\right] \\
&= \mathbb{E}\left[\alpha + \varepsilon_i \mid D_i = 0\right] \\
\mathbb{E}\left[Y_i\left(0\right) \mid D_i = 1\right] &= \mathbb{E}\left[Y_i - TE_i \mid D_i = 1\right] \\
&= \mathbb{E}\left[(\alpha + \beta + \varepsilon_i) - TE_i \mid D_i = 1\right] \\
&= \mathbb{E}\left[\alpha + \varepsilon_i \mid D_i = 1\right] + ATT - \mathbb{E}\left[TE_i \mid D_i = 1\right] \\
&= \mathbb{E}\left[\alpha + \varepsilon_i \mid D_i = 1\right] \\
\implies \mathbb{E}\left[Y_i\left(0\right) \mid D_i = 1\right] - \mathbb{E}\left[Y_i\left(0\right) \mid D_i = 0\right] &= \mathbb{E}\left[\alpha + \varepsilon_i \mid D_i = 1\right] - \mathbb{E}\left[\alpha + \varepsilon_i \mid D_i = 0\right] \\
&= \mathbb{E}\left[\varepsilon_i \mid D_i = 1\right] - \mathbb{E}\left[\varepsilon_i \mid D_i = 0\right]
\end{aligned}
$$

Thus, the selection bias term will be equal to $\mathbb{E}\left[\varepsilon_i \mid D_i = 1\right] - \mathbb{E}\left[\varepsilon_i \mid D_i = 0\right]$. This term will be zero if and only if the error term is uncorrelated with the treatment.